Evaluating UAV Coverage Plans Beyond Path Length Using Embedded-Deployable Nonlinear Model Predictive Control

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***Abstract*—Coverage path planners for agricultural UAVs are commonly ranked by path length. We test, in closed-loop simulation, how well length predicts the energy a multirotor spends on a boustrophedon plan flown with a rest-to-rest speed profile. On a 20×14 m field, two exclusion zones change the plan length by -4.5 to $+3.4\%$ but raise its energy by 8.3–13.6% (median 11.1%) over 299 generic positions (8.0% at the reference one), because the extra cells add stops and transits; planning energy is essentially hover power times flight time. A larger value, 22.6%, arises only from a degenerate decomposition that occurs when scan rows lie exactly on inflated obstacle edges. Over 600 random field, exclusion and sweep-direction combinations the median increases are 3.2% in length and 8.8% in energy. Flying the plans in closed loop, with a C-code-generated NMPC (`acados`) and PID/LQR baselines under Ornstein–Uhlenbeck force disturbances, preserves the planning-model ranking in the cases tested. The NMPC reduces tracking RMSE by 49–92% over the baselines; an ablation attributes 35–42% of that margin to its reference preview; given the reference acceleration as feedforward, PID tracks better and LQR worse than the NMPC without preview. Every controller comes closer to the no-fly polygons than the planned 0.5 m clearance, even without disturbance, and executed clearance, which the planning metrics do not show, can differ between traversal orders of equal planned cost.**

***Index Terms*—coverage path planning, model predictive control, energy-aware planning, software-in-the-loop, precision agriculture, benchmarking**

I. Introduction

Coverage path planning for agricultural UAVs is a mature field, and its results are often reported as path length, number of passes, or covered area [1], [2]. These are properties of a *curve*. What a battery pays for is a *trajectory*: a multirotor flying a boustrophedon sweep with a rest-to-rest profile decelerates at every turn and traverses whatever transit the cell structure imposes. That length is a poor proxy for energy is known [3], [4]; this paper asks how the gap behaves for a stop-and-turn agricultural sweep with exclusion zones, how much it depends on the geometry, and whether it survives closed-loop execution. Answering it requires a planner, a plant, a controller, a disturbance model and an energy model that are consistent with one another; we build that loop in software. Our contributions are:

1) **A closed-loop coverage benchmark** in which plans are scored both by a planning energy model and by the energy and clearance of the trajectory a controller actually flies.
2) **A geometric sensitivity study of the length–energy gap** over obstacle positions and random fields, including a degenerate case that inflates the reference value.
3) **An executed-safety measurement**: planned clearance is not executed clearance, even without disturbance.
4) **A paired evaluation** of NMPC against PID and LQR baselines over 360 missions, with an ablation that separates control law from reference information, and a delay sweep.

We claim no new control theory and no new planner: the predictive controller is reused from prior work, and the planner is a standard boustrophedon decomposition with Held–Karp cell ordering.

II. Related Work

Coverage planning. Galceran and Carreras [1] survey the field; boustrophedon cellular decomposition [5] handles non-convex regions, and Huang [6] shows that for convex polygons the sweep direction minimizing passes is parallel to an edge. Agricultural guidance tracks [2] and wind-aware survey decomposition [7] extend this line. Bähnemann *et al.* [4] pose cell ordering as a generalized TSP with rest-to-rest time costs, and Datsko *et al.* [3] plan multi-UAV coverage with an energy objective and optimal flight speed, validated in flight. We do not propose a planner; we measure how a length ranking and a planned clearance hold up against a closed-loop plant.

Embedded predictive control. CasADi [8] and `acados` [9] with HPIPM [10] and real-time iteration [11] make nonlinear MPC tractable at 100 Hz. We use this stack as a tool.

Simulation and sensing. Gazebo [12] with ROS 2 [13] is the standard open stack. NDVI [14] and NDWI [15] are canonical vegetation indices; infected potato canopy shows elevated visible and depressed NIR reflectance [16]; thermal CWSI follows Idso *et al.* [17].

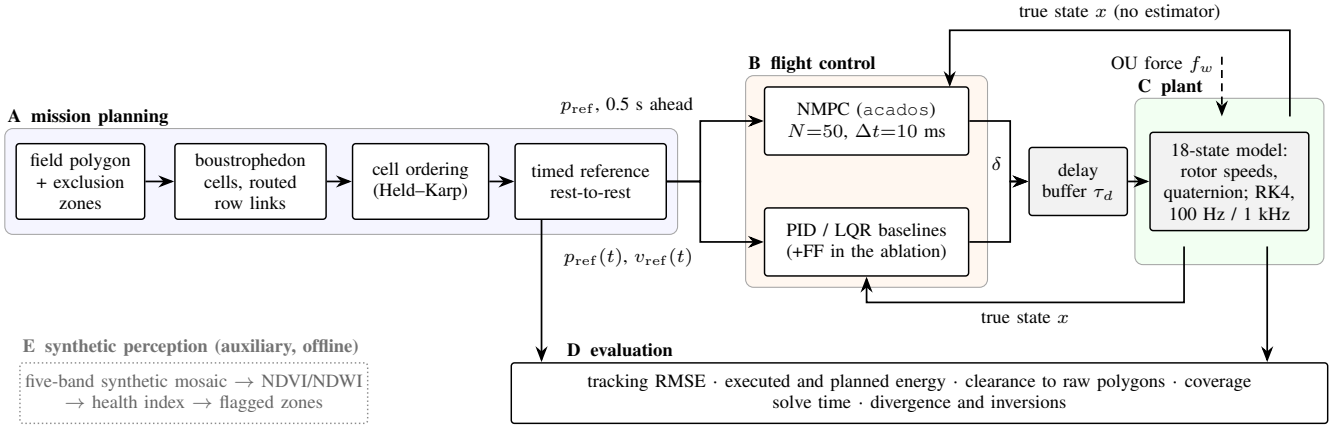


Fig. 1. Evaluation architecture used for every closed-loop result, run in Python (no Gazebo or ROS 2 in the loop). (A) The planner turns the field and exclusion zones into a timed rest-to-rest reference. (B) The NMPC receives the reference over its 0.5 s horizon; PID and LQR receive the current position and velocity reference (plus the reference acceleration in the feedforward ablation). Both act on the true state. (C) Rotor commands δ pass through an optional delay buffer (delay sweep) to the 18-state plant, integrated with RK4 at 100 Hz or 1 kHz under a world-frame Ornstein–Uhlenbeck force. (D) Metrics. (E) The synthetic perception benchmark is separate and offline.

III. Method

A. Vehicle model and disturbance

The plant is the 18-state model $x = [\,p^\top, v^\top, q^\top, \omega_b^\top, \varpi^\top, V_b\,]^\top$ with position p, world-frame (ENU) velocity v, attitude quaternion q, body rates ω_b, four rotor speeds ϖ and battery voltage V_b. Rotor i produces thrust $k_f\varpi_i^2$ and reactive torque $k_m\varpi_i^2$; the translational dynamics are

$$m\dot{v} = R(q)\,\textstyle\sum_i k_f\varpi_i^2\, e_3 - mge_3 - c_d\|v\|v + f_w, \quad (1)$$

with $m = 0.80$ kg, $k_f = 2.3 \times 10^{-5}$ N·s^2, $k_m = 7.0 \times 10^{-7}$ N·m·s^2 and $c_d = 0.15$ kg/m. Rotor speeds follow first-order actuator dynamics driven by normalized commands $\delta \in [0,1]^4$; V_b discharges with consumed current.

The disturbance f_w is a *world-frame* external force, not an airspeed: an Ornstein–Uhlenbeck process $\mathrm{d}f_w = -\theta f_w\,\mathrm{d}t + \sigma\,\mathrm{d}W$ with $\theta = 1$ s^{-1} (stationary standard deviation $\sigma/\sqrt{2\theta}$ per axis), plus, for $\sigma > 0$ only, a constant per-episode bias of 0.6 N whose direction is drawn per seed. The drag in (1) uses ground velocity, so the model has no relative-airspeed coupling. We keep it because the controller was tuned against it; Section VI states what this costs.

B. Coverage planning

The field is expressed in a local ENU frame; the sweep direction minimizes the perpendicular extent of the hull [6], and exclusion zones are inflated by a 0.5 m clearance and subtracted from the free area. Scan rows intersect the free area in spans; spans on consecutive rows that overlap belong to one cell, and a change in span count marks a critical point where cells close and open [5]. Three details matter for correctness: (i) when a row lies exactly on an inflated obstacle edge, the free area (a closed set) meets it in collinear spans that *touch*; they are merged into one span, since otherwise they read as critical points, create spurious cells and force stops mid-row; (ii) the link from one row to the next inside a cell, not only the transit between cells, is routed around the inflated obstacles (on a visibility graph, by Dijkstra); (iii) if the free area is disconnected, the plan is refused rather than joined by a segment through a no-fly zone. Each is covered by a regression test.

Cell ordering is an open-path TSP over cells and two traversal directions with *routed* transit lengths as edge costs, solved exactly by Held–Karp over two of the four entry corners per cell; it minimizes transit length, not energy or time. The planner emits a timed reference $t \mapsto (x, y, z, \psi)$ with a trapezoidal speed profile (2 m/s cruise) applied *per segment*, so the vehicle comes to rest at every segment end.

C. NMPC and baselines

The OCP is formulated in CasADi and exported to C by `acados`: $N = 50$ shooting nodes at $\Delta t = 10$ ms, Gauss–Newton Hessians, explicit Runge–Kutta integration, real-time iteration and partially condensed HPIPM. The attitude term is $\|\mathrm{vec}(q_{\mathrm{ref}}^* \otimes q)\|^2 = \sin^2(\alpha/2)$ for a relative rotation α: invariant to the quaternion sign, but not the squared geodesic distance. In the Monte-Carlo campaigns the NMPC receives the reference over its whole horizon with a fixed yaw, while the PID and LQR baselines, reused unchanged from the same prior work, act on the current reference only; the ROS 2 node instead extrapolates the reference and uses a measured-yaw governor. The comparison is therefore not information-equivalent. We write *embedded-deployable* because the artefact is the generated C code; no embedded processor executes it here, and we make no on-target timing claim.

D. Energy model

The planning model uses momentum theory with induced, profile and parasite power,

$$P_{\mathrm{mech}}(V) = k_i T v_i(V) + P_{p0}\big(1 + 4.65\mu^2\big) + c_d V^3, \quad (2)$$

TABLE I
COVERAGE COST, 20 × 14 M PLOT, 3 M SPACING (PLANNING MODEL)

	No exclusions	Exclusions (exact)	Exclusions (greedy)
Cells	1	6	6
Path length [m]	112.0	114.4	114.4
transit [m]	—	20.4	20.4
Mission time [s]	65.0	70.2	70.2
Free area [m^2]	280.0	234.0	234.0
Energy [Wh]	1.483	1.602	1.602
Energy/area [mWh m^{-2}]	5.30	6.84	6.84
Field coverage [%]	100.0	95.5	95.5
Accessible coverage [%]	100.0	99.5	99.5

with $v_i = v_h^2/\sqrt{V^2 + v_i^2}$, advance ratio $\mu = V/(\Omega R)$ and the same c_d as (1); the effective disc area is calibrated so that (2) reproduces the plant's hover power (82.0 W electrical). At 2 m/s it gives 82.2 W, so planning energy is, within 0.2%, hover power times mission time. For closed-loop runs we integrate $P = \sum_i k_m \varpi_i^3/\eta$ from the simulated rotor speeds.

E. Synthetic multispectral layer

A procedurally generated five-band reflectance mosaic (0.05 m) with canopy, soil and late-blight signatures [16] feeds NDVI [14] and NDWI [15], fused into a health index $H = 0.65\widehat{\mathrm{NDVI}} + 0.35\widehat{\mathrm{NDWI}}$ that is thresholded at τ. Each trial randomizes foci, rows, illumination, noise and severity, and τ is calibrated on trials disjoint from the test set.

IV. EXPERIMENTAL DESIGN

All missions cover a 20 × 14 m plot at 7 m AGL, 2 m/s and 3 m track spacing, unless stated otherwise. **Geometric sensitivity.** With the planning model only: the reference field with both obstacles shifted rigidly in y over ± 1.5 m in 1 cm steps (301 positions); and 200 random rectangular fields (15–60 m, aspect 1–3) with 1–3 exclusions (half of them rotated), each planned along three sweep directions (600 plans). Every relative change is computed per scenario against the same field without exclusions. **Monte Carlo.** PID, LQR and NMPC × $\sigma \in \{0, 0.3, 0.55, 0.8\}$ N·s$^{-1/2}$ × 30 seeds = 360 episodes on the exclusion-free field, plant at 100 Hz; seeds fix the OU realization and the bias direction. **Closed loop with exclusions.** The three controllers on the exclusion plan, $\sigma \in \{0, 0.55, 0.8\}$, 10 seeds (90 episodes), recording planned and executed energy and the minimum distance from the executed track to the raw no-fly polygons. **Delay sweep.** NMPC at $\sigma = 0.55$ with uncompensated actuation delay $\tau_d \in \{0, 2, 4, 5, 6, 7, 8, 10\}$ ms, 10 seeds each, plant at 1 kHz; an episode diverges if the state leaves a finite range or the position error exceeds 25 m. **Perception.** 10 calibration trials to select τ, then 30 disjoint test trials.

V. RESULTS

A. Geometric cost understates dynamic cost — by a geometry-dependent amount

On the reference field as placed (Table I, Fig. 2a,b) the exclusions lengthen the plan by 2.2% and raise its energy by 8.0%; per accessible area, which shrinks by 16.4%, the increase is 29.2%. Energy tracks mission time (+8.0%) because, at this speed, planning energy is hover power times time: each additional cell adds rest-to-rest segments and transit that cost time but little length.

A planner that does not merge touching spans gives, for the same field, 9 cells, +6.8% length and +22.6% energy. That value is a degeneracy: with integer obstacle coordinates, a 0.5 m clearance and rows at 1.5 m from the field edge and then every 3 m, three rows lie exactly on inflated obstacle edges, and such a planner splits each into touching spans that open spurious cells. Of the 305 positions swept with it (dashed in Fig. 2c), only the reference one produces it. Rows also meet inflated edges at $\Delta y = 1$ m, and even with span merging the reference position remains a special case: it is the unique minimum of the sweep. Over the other 299 positions the energy increase spans 8.3–13.6% (median 11.1%) while the length change spans −4.5 to +3.4%: in 33% of the positions the exclusions *shorten* the plan and still cost more energy. The cost is a step function of where the rows fall relative to the obstacles, so a single field cannot characterize it.

Over the 600 random plans the per-scenario median increases are 3.2% in length (IQR 1.2–5.3%) and 8.8% in energy (IQR 5.6–12.9%, 5th–95th percentile 2.4–23.0%). The median of the per-scenario ratio $\Delta E/\Delta L$ is 2.26 (IQR 1.54–3.47; over the 592 plans with $|\Delta L| > 0.1\%$, 75 of them negative), and the ratio of the medians is 2.74; the median ratio grows with the number of exclusion zones (1.77, 2.62 and 3.02 for one, two and three); in 13% of scenarios length falls while energy rises. Restricting to the 422 plans with accessible coverage ≥ 0.98 with and without exclusions gives a median energy increase of 8.6%. Pooled totals give +2.5% length and +7.8% energy. No plan entered an inflated obstacle; none was refused, because the generator keeps exclusions 1 m inside the field and 1.5 m apart.

Cell ordering. The 43% transit reduction that exact ordering shows on this field with the spurious cells also comes from them; without them, exact and greedy ordering give plans of identical length, time and energy, traversed in a different order. Over the random plans greedy ordering costs a median 0.7% more energy than exact ordering (> 1% in 46% of plans, > 5% in 8%), and can be up to 2.8% cheaper, since exact ordering minimizes transit length, not energy.

B. Closed-loop execution

Flying the exclusion plan preserves the planning ranking: without disturbance executed energy exceeds planned energy by 1.0–1.8%, and the exclusion premium measured in closed loop, paired by seed against the same controller and wind on the free field, is 7.4–9.1%, against 8.0% planned. With $\sigma > 0$ the persistent bias has a vertical component that shifts an episode's energy by several percent either way, which is why only paired comparisons are meaningful.

Planned clearance is not executed clearance (Table II, Fig. 3). Rows run along the inflated edges, so the reference itself sits at 0.5 m, and every controller drops below it in every episode,

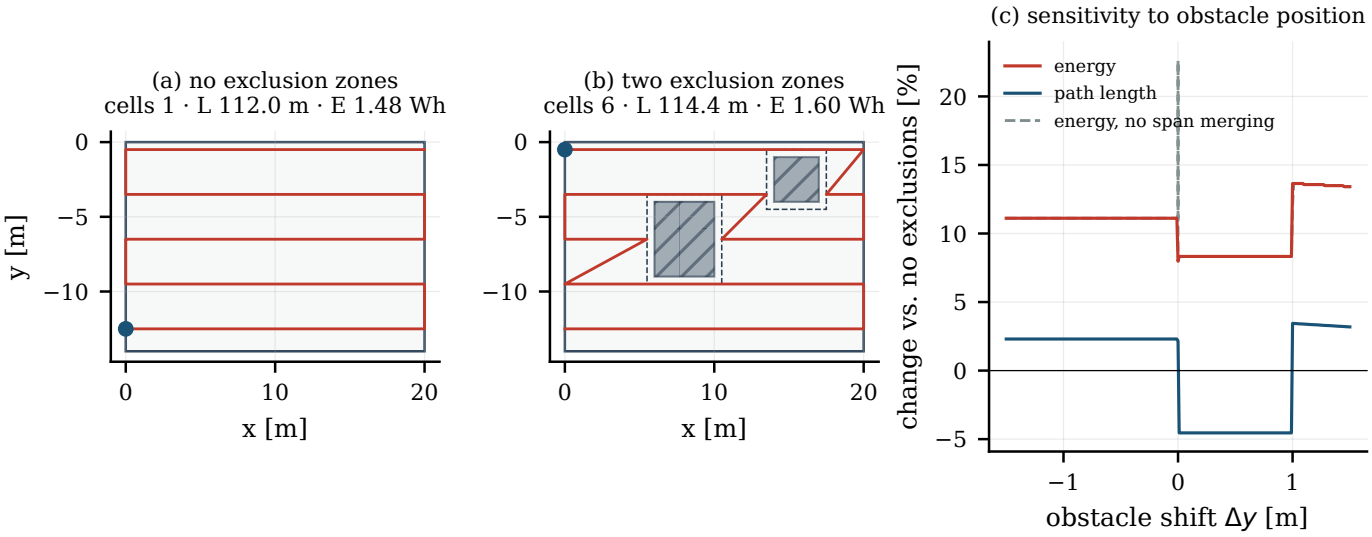


Fig. 2. (a,b) Plans without and with the two exclusion zones (hatched; dashed: inflated by the 0.5 m clearance); exact and greedy ordering give plans of equal length, time and energy here, flown in a different order. (c) Change in energy and path length caused by the exclusions when both obstacles are shifted by Δy; a planner without span merging (dashed grey) departs from ours only where scan rows lie on inflated obstacle edges ($\Delta y = 0$ and 1 m).

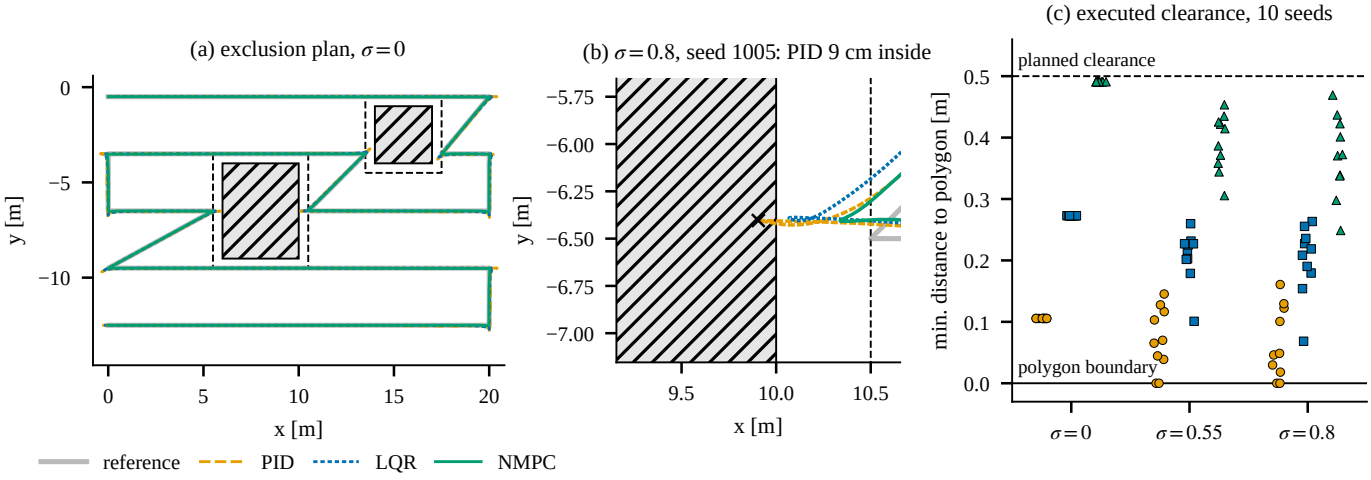


Fig. 3. Planned versus executed clearance on the exclusion plan (raw polygons hatched; dashed: inflated by the planned 0.5 m). (a) Reference and executed tracks without disturbance. (b) The deepest of the saved $\sigma = 0.8$ intrusions (exact order): PID overshoots a row end while braking and enters the polygon (×). (c) Minimum distance from each executed track to the raw polygons, 10 seeds per cell (at $\sigma = 0$ the seeds coincide). Tracks are the episodes of Table II, re-run with identical metrics.

even without disturbance. The NMPC keeps at least 0.249 m; LQR comes within 0.07 m; PID enters a no-fly polygon in 2 of 10 episodes at $\sigma = 0.55$ and at $\sigma = 0.8$, through overshoot when braking at row ends that face an obstacle. Flown in the greedy order, the same plan gives PID intrusions in 1 and 5 of 10 episodes and brings LQR within 0.05 m: the executed clearance may depend on the traversal order, which the planning metrics do not see (2 against 5 of 10 PID intrusions at $\sigma = 0.8$, Fisher $p = 0.35$: not resolved by 10 seeds). The PID–LQR difference is not statistically significant at this sample size; with reference feedforward neither baseline enters a polygon in the exact order (Sec. V-C).

C. Tracking: Monte Carlo against baselines

Table III reports the 360-mission campaign; a re-run with the versioned scripts reproduces Table III to three decimals. The NMPC reduces RMSE by 48.8–92.2% relative to the better linear baseline at every wind level, with no divergences. At $\sigma = 0$ the 30 episodes per controller are identical (no turbulence and no bias), so that row is a single run.

The energy differences between controllers are small but not null. Paired by seed over the 90 episodes with $\sigma > 0$, the NMPC uses -10.2 mWh (-0.67%; 95% CI -10.5 to -9.9) relative to PID (paired t-test $p < 10^{-6}$) and -0.4 mWh (-0.03%; CI -0.9 to 0.1; $p = 0.10$) relative to LQR, a difference whose sign changes with the wind level and between the deposited run

TABLE II
CLOSED LOOP ON THE EXCLUSION PLAN (EXACT ORDER): RMSE [M], MINIMUM DISTANCE FROM THE EXECUTED TRACK TO THE RAW NO-FLY POLYGONS [M] OVER 10 SEEDS, EPISODES BELOW THE PLANNED 0.5 M CLEARANCE AND EPISODES ENTERING A NO-FLY POLYGON

	σ	RMSE	min	median	< 0.5 m	inside
PID	0.00	0.213	0.106	0.106	10/10	0/10
PID	0.55	0.245	0.000	0.068	10/10	2/10
PID	0.80	0.272	0.000	0.047	10/10	2/10
LQR	0.00	0.187	0.273	0.273	10/10	0/10
LQR	0.55	0.242	0.101	0.221	10/10	0/10
LQR	0.80	0.267	0.068	0.214	10/10	0/10
NMPC	0.00	0.013	0.491	0.491	10/10	0/10
NMPC	0.55	0.100	0.306	0.401	10/10	0/10
NMPC	0.80	0.122	0.249	0.371	10/10	0/10

TABLE III
TRACKING RMSE [M], MEAN $\pm$ STD OVER 30 SEEDS PER CELL

σ	PID	LQR	NMPC	Δ vs best
0.00	0.181 ± 0.000	0.167 ± 0.000	$\mathbf{0.013 \pm 0.000}$	-92.2%
0.30	0.191 ± 0.005	0.209 ± 0.010	$\mathbf{0.083 \pm 0.007}$	-56.7%
0.55	0.212 ± 0.009	0.226 ± 0.011	$\mathbf{0.101 \pm 0.007}$	-52.4%
0.80	0.242 ± 0.012	0.251 ± 0.013	$\mathbf{0.124 \pm 0.008}$	-48.8%

and the re-run. All differences lie within a $\pm 1\%$ equivalence margin, so the energy effects of Table I belong to the plan, not to the controller tracking it.

How much of the margin is the controller? The NMPC sees the reference over its whole horizon; the baselines see only the current position and velocity reference. On the exclusion plan at $\sigma = 0.55$, giving PID and LQR the reference acceleration as feedforward lowers their RMSE from 0.245 and 0.242 to 0.145 and 0.188 m, and removing the preview from the NMPC (current reference extrapolated with its velocity, as in our ROS 2 node) raises its RMSE from 0.100 to 0.160 m, between the two feedforward baselines although it receives no reference acceleration. With feedforward no baseline enters a no-fly polygon in 30 episodes (PID without it: 4). The preview thus accounts for 35–42% of the NMPC's margin over the original baselines ($\sigma = 0.55$, 0.8); with feedforward, PID tracks better and LQR worse than the NMPC without preview in all 20 paired episodes, so the information is not matched exactly; without preview the NMPC also uses 1.9% more energy.

NMPC solve times, measured on a desktop CPU during the 4-way parallel re-run, average 1.33–2.76 ms per episode (median 99th percentile 2.91 ms) against the 10 ms budget, but the slowest solve exceeds it in 33 of 120 episodes (up to 32.9 ms); in the original campaign of Table III, whose load conditions were not recorded (means 3.05–4.39 ms), it did so in all 120. These are desktop figures under load, not embedded ones.

D. Actuation delay

Up to 2 ms the delay is invisible beside the wind-driven error; RMSE roughly doubles at 4 ms and is 9.2× nominal at 6 ms. At 5–6 ms no episode exceeds the 25 m divergence criterion,

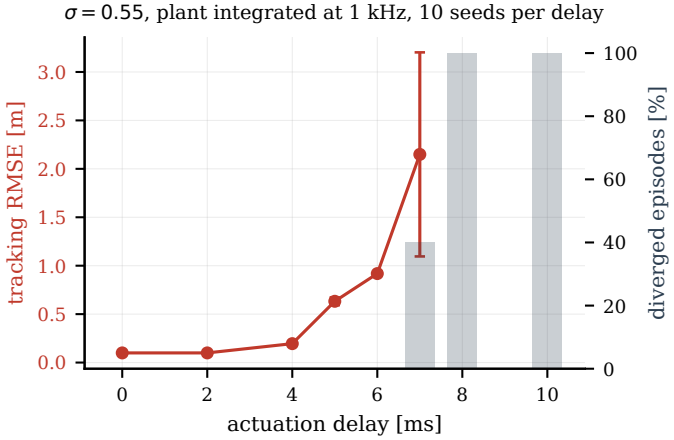


Fig. 4. Uncompensated actuation delay at $\sigma = 0.55$ (free field, plant at 1 kHz, 10 seeds per delay): RMSE of runs below the 25 m criterion and share of diverged episodes.

yet the vehicle inverts in all 10 episodes (3–35 inversions each; none at 4 ms): attitude control is lost between 4 and 5 ms. At 7 ms 4 of 10 episodes diverged in one run and 6 of 10 in a re-run with a newer solver build, which agree on only 4 of the 10 seeds; from 8 ms all episodes diverge, accompanied by thousands of solver failures. This is an empirical transition for this controller, fallback policy and scenario — not a stability margin — and at 7 ms the divergence rate is not resolved by 10 seeds.

E. Perception

NDVI separates healthy canopy (0.851) from diseased (0.428) and bare soil (0.091), NDWI 0.249 from -0.048. Over 30 test trials at the calibrated $\tau = 0.65$, flagged zones reach precision 0.683 ± 0.216, recall 0.641 ± 0.247 and $F_1 = 0.607 \pm 0.200$ (mean $\pm$ s.d.); τ lies close to a steep drop of the calibration curve, so these values are sensitive to it. The signatures and the ground truth share a generator, so this benchmark measures the chain from indices to zones, not detection of real disease. A single Gazebo Classic mission ($N = 1$, threshold chosen on it; in the repository) only shows that georeferencing closes end to end.

VI. DISCUSSION AND LIMITATIONS

Without the degeneracy, for a rest-to-rest sweep path length understates the energy cost of exclusion zones by a factor that varies with geometry (median 2.26 over random plans, with opposite signs in some configurations). Planners that score time or energy [3], [4], [18] already account for this; ours, which orders cells over two entry corners and minimizes transit length, is not competitive with them. What this study adds is the closed-loop side: the ranking survived execution, the planned clearance did not. Because planning energy here is hover power times flight time, the ranking is in effect a time ranking: faster cruise or smooth turns would change its magnitudes, which we did not evaluate.

The clearance a plan must reserve depends on the controller that flies it: in Table II the NMPC consumed up to half of the

0.5 m margin under disturbance, LQR nearly all of it, and PID all of it in some episodes, and plans of equal planned cost flown in different orders may consume it differently. A controller-specific inflation, or a planner scoring predicted tracking error at row ends, is the natural extension (not evaluated here). With control lost at 4–5 ms of delay and desktop means of 1.33–4.39 ms (maxima above 10 ms), embedded deployment needs a latency analysis we do not provide.

Results assume rest-to-rest segments (smooth turns were not evaluated) and a world-frame force without airspeed coupling (relative-airspeed turbulence is open). The NMPC has a perfect model, and the ablation matches reference information only approximately. The plan depends on the frame: rotating the field moves the energy increase from 6.0 to 20.7% over 72 angles, a ± 1 mm tolerance where rows meet inflated edges moves it from 8.0 to 17.7%, and rows can skip narrow strips (free-field coverage < 0.98 in 135 of 600 plans). All results are simulated with zero computation delay on a non-identified airframe (thrust up to $26\times$ weight), and both energies are model outputs (the planning model omits acceleration thrust; the closed-loop one uses a different rotor model), not measured battery consumption.

VII. Conclusion

Observed. For a rest-to-rest boustrophedon sweep, two exclusion zones on a 20×14 m field change the plan length by -4.5 to $+3.4\%$ but cost 8.3–13.6% more energy at generic positions (8.0% at the reference one), because they add stops and transits; a larger effect at the reference position, and a benefit of exact cell ordering there, come from a degenerate decomposition. *Within this benchmark*, over 600 random plans the median costs are 3.2% in length and 8.8% in energy, the planning ranking survives closed-loop flight in the cases tested, part of the NMPC's tracking advantage is reference preview, and no controller keeps the planned clearance, which may also depend on the traversal order. *Not shown*: energies measured on a battery, a physically identified airframe, smooth-turn profiles, or execution on embedded hardware. Planning that accounts for the tracking error of the controller that will fly the plan is the natural next step.

Reproducibility

The planner (with regression tests), plant, controller, Monte-Carlo harness, input data with provenance records and the scripts that regenerate the numbers, Tables I–II and the figures (`make_numbers.py`, `coverage_study.py`, `make_figures.py`, `fig_clearance.py`) are available at https://github.com/JoseSegundoM/uav_energy_coverage. Input data carry SHA-256 checksums and every episode records its seed; `make_numbers.py` regenerates the numbers and Tables I–II byte for byte, and Table III matches `mc_main.json`.

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